

Mimetic marine antifouling films based on fluorine-containing polymethacrylates



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ABSTRACT

Novel methacrylate copolymers containing catechol and trifluoromethyl pendant side groups were synthesized by free radical polymerization of *N*-(3,4-dihydroxyphenyl)ethyl methacrylamide (DMA) and 2,2,2-trifluoroethyl methacrylate (TFME) with α,α' -azobisisobutyronitrile (AIBN) as initiator. A series of copolymers with different content of TFME ranging from 3% to 95% were obtained by changing the molar ratio of DMA to TFME from 25:1 to 1:25. Fourier transform infrared (FT-IR) spectroscopy, gel permeation chromatography (GPC), thermal gravimetric analysis (TGA), differential scanning calorimetry (DSC) and scanning electron microscopy (SEM) were used to characterize the copolymers, which displayed a certain degree of hardness and outstanding thermostability reflected from their high glass transition temperatures. The copolymers could adhere to surfaces of glass, plastics and metals due to introduction of catechol groups as multivalent hydrogen bonding anchors. Water contact angle on the polymer films was up to 117.4°. Chemicals resistance test manifested that the polymer films possessed excellent resistance to water, salt, acid and alkali. Moreover, the polymer films displayed fair antifouling property and might be used as promising environmentally friendly marine antifouling coatings.

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1. Introduction

Mankind has been fighting against biofouling for more than 2000 years. Biofouling arises from the undesirable attachment and colonization of marine organisms, plants, algae [1], or animals on surfaces submerged in seawater. Serious adverse impact from biofouling has been recognized including impairment of the operation efficiency of ships and other oceanographic equipments or instruments, increased fuel consumption, corrosion, emission of gases such as CO_x, SO_x and NO_x into the atmosphere, and incursion of invasive species into marine habitats. Various approaches for the suppression of biofouling have been envisaged, including enzymatic degradation, sonication and chemical modification of the substrates. Although the self-polishing coatings based on tributyltin (TBT) or copper pyrrhione are effective on removing or reducing biofouling over the last few decades because of slow release of tin, copper or other toxins in time, their use has been restricted through legislation due to environmental concerns. Subsequently, development of non-toxic and environmentally benign

antifouling coatings has become an attractive and challenging field and a main trend in marine industry. Enormous antifouling materials were developed to alter the physical and/or chemical properties of the surface in order to prevent the accumulation of the organisms on the substrates. To date zwitterionic polymers [2–5], biodegradable hydrophilic or amphiphilic polymers [6–8], bioinspired engineered topographies [9,10], (nano-structured) organic-inorganic hybrid systems [11–13], low surface energy elastomers specifically silicone or fluoropolymers [14,15], agricultural and natural products [16,17], and self-assembly of a low molecular weight peptide [18] have been used for preparation of antifouling coatings. Shark skin effect has also been highly valued for fabrication of biomimetic anti-biofouling coatings [19].

Poly(tetrafluoroethylene) (PTFE) is the benchmark low surface energy material but is far from efficient as antifouling material due to its low oil repellency and to its microcrystalline surface structure [20]. Therefore considerable research efforts have been directed toward the development of alternative low surface energy polymeric materials with good film-forming property. In this respect catechol 3,4-dihydroxyphenylalanine (DOPA), as one of the most important functionalities of adhesive proteins in mussels, has exerted a very important role because it can strongly adhere a wide variety of organic and inorganic solid surfaces including PTFE

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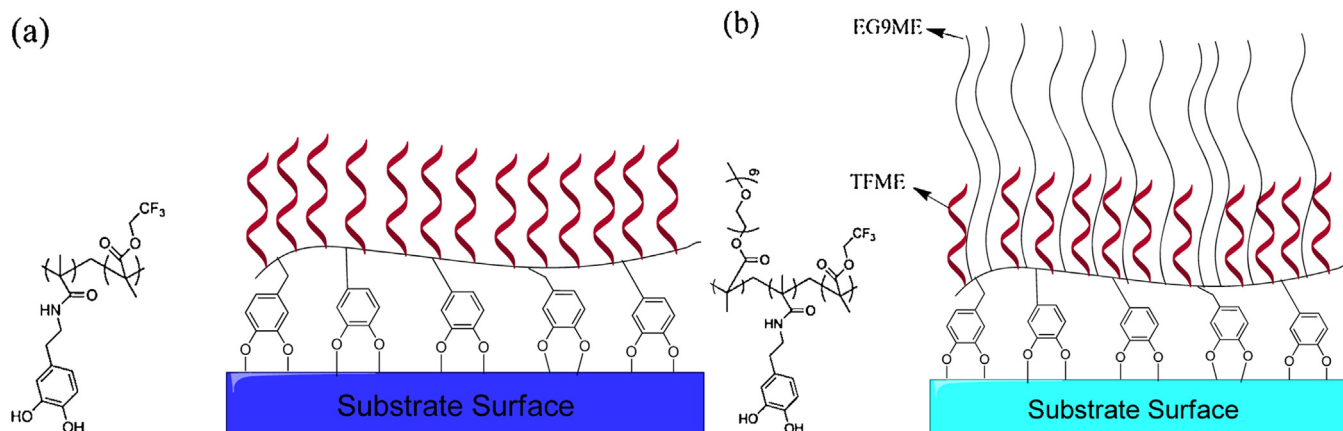


Fig. 1. Molecular structural model of copolymer (a) and ternary polymer (b).

surfaces and the adhesion properties are present even in water [21,22]. Owing to this high adhesion property in water, DOPA-functionalized poly(ethylene oxide) has been applied to antifouling surfaces in the biotechnology [4]. Messersmith and colleagues designed and synthesized DOPA-containing peptide-PEG conjugates for antifouling purposes, reasoning that DOPA-containing peptides would provide robust and versatile anchors for the immobilization of antifouling polymers onto metal, metal oxide, and polymer surfaces [23,24]. Recently DOPA was used for deposition of antifouling coatings on stainless steel via catechol-amine reaction [25].

Herein, we report synthesis, characterization and property study on biodegradable copolymers containing both catechol groups able to anchor onto the substrates to form coatings and fluorinated alkyl chains to impart the surfaces of the copolymers with low surface free energy. Thus DMA was combined with TFME to produce a new type of antifouling copolymers poly(DMA-co-TFME) as showed in Fig. 1(a) and a ternary polymer poly(DMA-co-TFME-co-EG9ME) (DFE) by introduction of the third monomer poly(ethylene glycol) methyl ether methacrylate (EG9ME) as showed in Fig. 1(b). Different copolymers could be obtained by varying the feed ratio of DMA to TFME. Specifically, the antifouling polymers described here not only can be designed to promote adhesion between two dissimilar surfaces, they can also be designed to prevent adhesion of undesirable particles, i.e. cells, proteins, and bacteria. For example, a high DMA content may be required for a coating that promotes adhesion, as enough DMA is needed to coat the support substrates as well as to promote adhesion to a second substrate. On the other hand, a lower DMA content may be needed for an antifouling coating since excessive DMA could promote unwanted adhesion. In this work the starting materials are inexpensive and the polymers containing short fluorinated alkyl chains are easy to degrade.

2. Experimental

2.1. Materials

Methacryloyl chloride (AR), 2,2,2-trifluoroethanol (AR) and poly(ethylene glycol) methyl ether methacrylate (EG9ME, $M_n \sim 475$) were purchased from Shanghai Energy Chemicals Co., Ltd. and were used as received. Dopamine hydrochloride was obtained from Beijing HWRK Chemical Co., Ltd. Triethylamine (AR), methacrylic acid (AR), 4-dimethylaminopyridine (DMAP, AR), thionyl chloride (AR), phenothiazine (AR) and other reagents were supplied by China National Pharmaceutical Group Chemical Reagent co., Ltd. AIBN (AR) was purchased from Aldrich and was

used after purification by recrystallization in ethanol. Methanol and *N,N*-dimethylformamide (DMF) were dried over CaH₂ and distilled just prior to use. Other reagents and solvents obtained from commercial suppliers were used without further purification.

2.2. Instrumentation and characterization

FT-IR spectra were collected on an Avatar 380 spectrophotometer with a computer running PE-Spectrum v5.3 software. Liquid samples were presented on KBr disks and solid samples were mixed with KBr pellets to give a tablet. Then 32 scans were recorded over the range 4000–400 cm⁻¹ at a resolution of 4 cm⁻¹ at intervals of 1 cm⁻¹.

Nuclear magnetic resonance (NMR) spectra were measured on a Bruker Avance-400 spectrometer with tetramethylsilane (TMS) as an internal reference (*s*=singlet, *d*=doublet, *t*=triplet, *m*=multiplet, *br*=broad). The chemical shifts are reported on the ppm scale for ¹H NMR with a resonance frequency of 400 MHz.

GPC results were obtained on a Waters BI-MWA instrument with tetrahydrofuran (THF) or DMF as eluent at a flow rate of 1 mL min⁻¹ using polystyrene as a standard. Samples of polymer DF25-1 were dissolved in THF while those of other polymers (DF1-1, DF1-2.5 and DF3-1) were dissolved in DMF.

TGA was carried out by NETZSCH TG209F1. Samples (no more than 5.0 mg) in ceramic crucible were heated from room temperature to 900 °C with a heating rate of 10 K min⁻¹ under a nitrogen atmosphere (flow rate: 40 mL min⁻¹). All analyses were repeated to ensure consistency.

DSC was performed on NETZSCH DSC204F1 under a nitrogen atmosphere (flow rate: 20 mL min⁻¹). Samples (no more than 5.0 mg) in hermetically-sealed aluminium pans were heated at 10 K min⁻¹ from room temperature to 200 °C just below the degradation temperature of the sample (previously determined by TGA).

SEM images were obtained on a TM-1000 with Hitachi desktop scanning electron microscope at 15 kV.

Contact angle measurements were carried out on DSA30 contact angle goniometer (polymers drop-casted from ethyl acetate (EtOAc) solution onto a glass surface, triplicates for each sample with standard variation below 5%, static contact angle of deionized water against the surfaces). Deionized water was chosen as testing liquid because advancing water contact angle analysis is a rapid and convenient means of determining if a coating was successfully applied. Contact angle values of the polymer films were determined at 25 °C by using a goniometer interfaced to image capture software via injecting a 3 μL drop of the test liquid onto the surface. The contact angle values were then quantified using image analysis software Image. The average contact angle value for each

sample was obtained based on 6 contact angle values on different locations.

The morphology of the membrane surface was observed by using a BA310 biological microscope from Motic China Group Co., Ltd. The optical microscope was equipped with a digital camera, CCIS infinity chromatic aberration correction system and 30° hinged trinocular head [F.N.20]–binocular.

2.3. Synthesis of monomers and polymers

2.3.1. Synthesis of DMA

To a nitrogen purged 100 mL three-neck flask were added 3,4-dihydroxyphenylalanine hydrochloride (4.0 g, 21 mmol) and methanol (37 mL). The mixture was degassed with nitrogen for 30 min and triethylamine (3.0 mL, 2.18 g, 22 mmol) was added. Then methacryloyl chloride (2.7 mL, 2.92 g, 27 mmol) in THF (2.5 mL) was added to the reaction mixture under ice bath by a dropping funnel. At last additional triethylamine (1.8 mL, 1.31 g, 13 mmol) was added. The reaction was monitored by thin layer chromatography (TLC). The mixture was stirred for 6 h at room temperature. Methanol was removed by rotary evaporation and the residue was extracted with ethyl acetate. The organic layer was washed with 1 N HCl and brine, and dried over anhydrous MgSO_4 . The solution was concentrated by rotary evaporation and the crude product was obtained, which was recrystallized in ethyl acetate at low temperature (-21°C) and dried in vacuo to afford 2.53 g (11 mmol, yield 54.3%) DMA as white powder. Mp $136\text{--}138^\circ\text{C}$. ^1H NMR (400 MHz, DMSO, δ): 8.77 (s, 1H, OH), 8.65 (s, 1H, OH), 7.96 (s, 1H, NHCO), 6.63 (d, $J=7.9$ Hz, 1H, Ar-H), 6.57 (s, 1H, Ar-H), 6.43 (d, $J=7.9$ Hz, 1H, Ar-H), 5.62 (s, 1H, HC=C), 5.30 (s, 1H, HC=C), 3.22 (t, $J=8.0$ Hz, 2H, NCH_2), 2.55 (t, $J=8.0$ Hz, 2H, ArCH_2), 1.84 (s, 3H, CH_3). FT-IR (KBr): ν 3356, 3174, 2925, 1646, 1589, 1461, 1360, 1268, 1196, 961, 787, 646 cm^{-1} .

2.3.2. Synthesis of TFME

Thionyl chloride (7.38 g, 66 mmol) and phenothiazine (0.05 g) were added to a 100 mL three-neck flask. The mixture was stirred evenly with a magnetic stirrer. Then methacrylic acid (4.3 g, 50 mmol) was dropped slowly into the reaction mixture. After addition the mixture was heated to 65°C and refluxed for 2 h. DMAP (0.025 g) was dissolved in 2,2,2-trifluoroethanol (5.5 g, 55 mmol) and added into above reaction mixture dropwise by a constant pressure dropping funnel. The reaction was performed at 65°C for 4 h with continuous stirring. At this point, the liquid of the mixture was faint yellow, cooled to room temperature and neutralized with 20 wt% sodium hydroxide solution, then layered after still standing. The reaction liquid was filtered to get yellow liquid which was washed with brine (5 mL) for three times, dried over anhydrous sodium sulfate and distilled under reduced pressure to collect bp $31^\circ\text{C}/101.3$ Pa fraction. At last 4.0 g of colorless and transparent liquid TFME was obtained in 47% yield. ^1H NMR (400 MHz, CDCl_3 , δ): 6.23 (s, 1H, HC=C), 5.71 (s, 1H, HC=C), 4.54 (q, $J=8.0$ Hz, 2H, CH_2CF_3), 1.98 (s, 3H, CH_3). ^{19}F NMR (377 MHz, CDCl_3 , δ): -73.93 (t, $J=8.0$ Hz, CF_3). FT-IR (KBr): ν 2920, 1740, 1632, 1280, 1149, 970, 659 cm^{-1} .

2.3.3. Preparation of poly(DMA-co-TFME)

Into a polymerization tube were added DMA (0.22 g, 1 mmol) dissolved in DMF (2 mL) and AIBN (4.5 mg, 0.03 mmol, 1.1 wt%). The solution was bubbled for 30 min with dry nitrogen to remove the oxygen and finally the tube was charged with dry nitrogen. TFME (0.185 g, 1.1 mmol) was added into the tube with a syringe. The polymerization was conducted with a magnetic stirrer at 65°C for 24 h. The reaction mixture was cooled to room temperature and diethyl ether (20 mL) was added. The precipitate was collected and purified by reprecipitation into a large amount of diethyl ether.

Finally, the copolymer was dried under reduced pressure to give a white solid (1.76 g, 80.1% conversion). A series of different fluoro-copolymers were synthesized by changing the feed molar ratio of reaction monomer DMA to TFME.

2.3.4. Preparation of the ternary polymer

poly(DMA-co-TFME-co-EG9ME) (DFE)

Into a polymerization tube were added DMA (0.22 g, 1 mmol) and EG9ME (1.0 g, 2 mmol) dissolved in DMF (4 mL) and AIBN (0.04 g, 0.24 mmol, 1.5 wt%). The mixture was bubbled for 30 min with dry nitrogen to remove the oxygen and finally the tube was charged with dry nitrogen. 1.344 g (8 mmol) TFME was added into the tube with a 2 mL syringe. The polymerization was conducted by using a magnetic stirrer at 65°C for 20 h. The reaction mixture was allowed to cool to room temperature and diethyl ether (80 mL) was added. The precipitate was collected and purified by reprecipitation into a large amount of diethyl ether. After washing for several times the copolymer DFE was dried under reduced pressure to give a white viscous solid (2.19 g, 85.4% conversion).

2.3.5. Preparation of poly(DMA-co-TFME) films

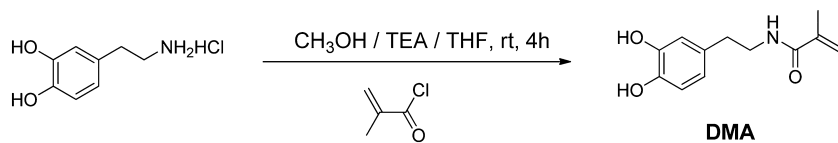
Different polymer solutions were prepared by dissolving the polymers in EtOAc or DMF and were drop-casted onto clean glass, metal and plastic substrates, respectively, by natural drying and then drying in a vacuum oven (60°C) overnight to remove the solvent. Crisp and transparent films of poly(DMA-co-TFME) were obtained after thorough drying.

3. Results and discussions

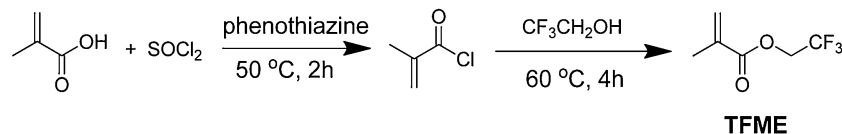
3.1. Synthesis of monomers and functional copolymers

DMA containing a catechol group was prepared by the amidation reaction between methacryloyl chloride and 3,4-dihydroxyphenylalanine hydrochloride, as showed in [Scheme 1](#). The resultant DMA was soluble in EtOAc but almost insoluble in water. Spectral data for the monomer were in good accordance with those reported in the literature [\[26\]](#). Another monomer TFME was prepared by esterification of methacrylic acid with 2,2,2-trifluoroethyl alcohol as showed in [Scheme 2](#) by a one-pot two-step procedure. The obtained TFME turned as colorless and transparent liquid with slight odor of stimulating fragrance and was insoluble in water. ^1H NMR, ^{19}F NMR and FT-IR spectral data for the monomer were measured to verify its structure [\[27\]](#). Then DMA was copolymerized with TFME through free radical polymerization in DMF as showed in [Scheme 3](#). The polymerization reaction was conducted at 65°C with AIBN as initiator due to induced decomposition nonoccurrence compared to organic peroxides initiators and its suitable half-life period (5 h at 70°C , coordinating with the reaction time). The suitable mass fraction of AIBN in the comonomers is about 1–5 wt% and was set at 1.1 wt% in this study.

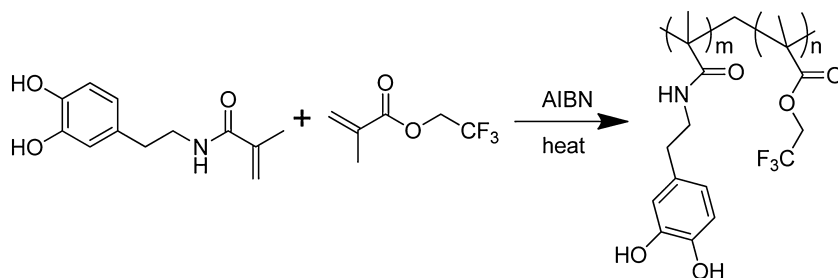
Polymerization was carried out without protection of the reactive DMA side chain, which reduces synthetic steps and allows the polymers to be prepared in a higher yield, as showed in [Scheme 3](#). Although phenolic compounds are known to be inhibitors and radical scavengers, the removal of atmospheric oxygen allowed us to synthesize poly(DMA-co-TFME) copolymers of high molecular weight [\[28–30\]](#). A series of poly(DMA-co-TFME) polymers were obtained with the content of TFME varying from 3 wt% to 95 wt%. The prepared polymers could be purified by precipitation into diethyl ether but it was difficult to separate the polymer from the reaction mixture even by use of ultrafiltration method due to the existence of residual DMA in case that the solubility of DMA was similar to that of the polymer. Nevertheless, the content of residual DMA in the polymer is too tiny to affect the subsequent process of coating formation.



Scheme 1. Synthetic route to DMA.



Scheme 2. Synthetic route to TFME.



Scheme 3. Synthetic scheme for poly(DMA-co-TFME).

Table 1

The dose of the used materials and reaction conditions for synthesis of antifouling polymers.

Polymer	Monomer feed molar ratio DMA:TFME	TFME wt%	AIBNwt%	DMF /mL	T/°C	t/(h)	Conv./%
DF25-1	25:1	3.0	1.1	2	65	32	77.1
DF3-1	3:1	20	1.1	2	65	32	39.2
DF1-1	1:1	43	1.1	2	65	20	77.3
DF1-2.5	1:2.5	65	1.1	4	65	30	82.6
DF1-10	1:10	88	1.1	2.5	65	30	80.1
DF1-25	1:25	95	1.1	2	65	18	83.2

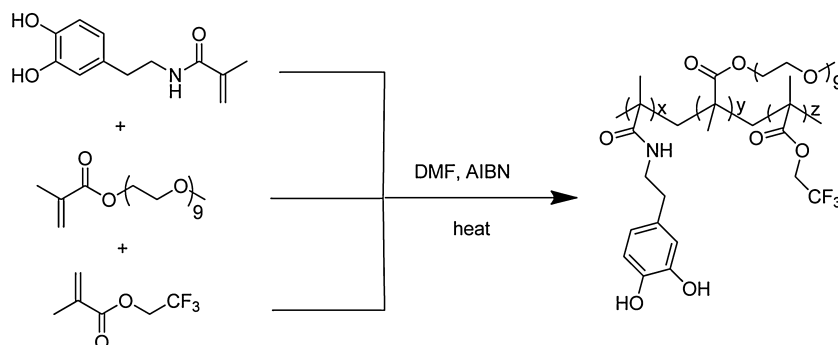
Conv./% means conversion ratio and the conversion was calculated based on the equation: $\text{Conv.}/\% = W_p/W_M \times 100\%$, where W_p is the weight of the polymer after drying to the constant weight and W_M represents the weight of the monomer.

T: reaction temperature.

t: reaction time.

By altering DMA:TFME feed ratio, it is possible to control the molecular weight and the composition of the resulting polymers. As shown in the Table 1, DMA:TFME feed molar ratio varied from 25:1 to 1:25, resulting in polymers with TFME content ranging from 3 wt% to 95 wt%. The DMA segment in a poly(DMA-co-TFME) polymer can be used to control the adhesive nature of the polymer, as it has been demonstrated that higher DMA content correlates to

stronger adhesive strengths. Additionally, TFME has a good stain resistance and hydrophobicity, which has been used in many products. Hydrophobic segments such as trifluoromethyl group can be incorporated to increase the stiffness of the polymer backbone, which can result in aggregation of these hydrophobic regions in an aqueous media as well as increasing the mechanical strength of the formed polymers film. Different types of chemical linkages can



Scheme 4. Synthetic scheme for poly(DMA-co-TFME-co-EG9ME).

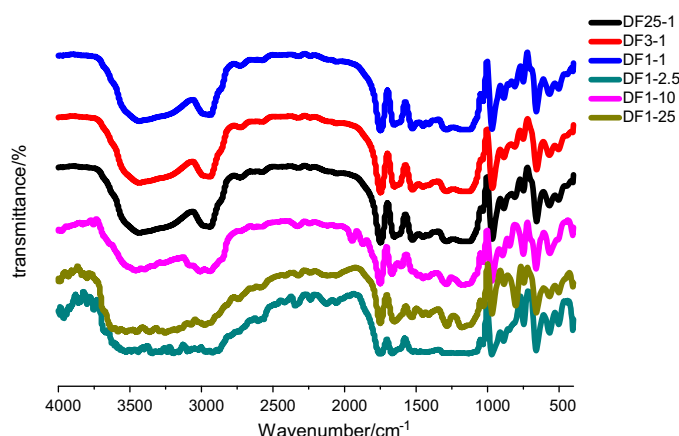


Fig. 2. FT-IR spectra of poly(DMA-co-TFME) polymers.

be used to control the stability and the rate of degradation of the polymer backbone and the linkage used in this study was stable C–C bond. The presence of methyl group in the methacrylate moiety increases the hydrophobicity and the stiffness of the polymer backbone and reduces the solubility of the polymers as opposed to a hydrogen atom. For example, the prepared poly(DMA-co-TFME) polymers are water insoluble and not readily soluble in diethyl ether but are soluble in a number of organic solvents such as DMF, chloroform, THF and EtOAc. The higher hydrophobicity ascribed to catechol unit and trifluoromethyl groups could translate into better compatibility with oil-based paints, of significance for the ultimate goal of the study to develop effective marine antifouling coatings. Moreover, EG9ME was introduced into the polymer chain to form a ternary polymer as showed in Scheme 4 due to superior antifouling feature of poly(ethylene glycol) (PEG) segment, which also improved the flexibility and the film-forming of the polymer.

3.2. Structural characterization of poly(DMA-co-TFME) polymers

3.2.1. FT-IR analysis of poly(DMA-co-TFME) polymers

FT-IR spectra of poly(DMA-co-TFME) copolymers were measured and showed in Fig. 2. All of the copolymers exhibited a characteristic absorption band of the carbonyl group at around 1750 cm^{-1} . The absorption peaks at ca. 1640 and 1500 cm^{-1} were ascribed to aromatic ring stretching vibration. The band at around 1280 cm^{-1} was attributed to C–O and C–F bond stretching vibration. The wagging vibration of C–F bond was observed at ca. 660 cm^{-1} . The broad strong absorption peak at ca. 3400 cm^{-1} was resulted from the stretching vibration absorption of ArO–H overlapping with that of N–H in the secondary amino group. The band

centered at 2936 cm^{-1} was attributed to saturated C–H stretching in methyl group. It could be seen that the bands of phenolic hydroxyl and amino group for DF1-2.5 and DF1-25 were wider than those for other poly(DMA-co-TFME) copolymers due to the hydrogen bonding effect [31]. And the rocking vibration of C–H bond in CH_3 group occurred at around 970 cm^{-1} . These results were indicative that the copolymers had been synthesized successfully and different feed molar ratio between monomers did not significantly alter the structures of the copolymers.

3.2.2. GPC measurement of the copolymers

Number-average molecular weight (M_n) and mass-average molecular weight (M_w) of the copolymers were measured by conventional GPC. Table 2 showed the molecular weights and polydispersity index ($\text{PDI} = M_w/M_n$) of the polymers, together with their thermal-decomposition temperature (T_d) and glass transition temperature (T_g). It can be seen that the dispersity indices of most polymers are relatively narrow even for those with a high degree of polymerization of polymethacrylate backbone. When the content of TFME in the copolymer reaches 43 wt%, the highest molecular weight ($M_w = 1.32 \times 10^5$) was achieved for DF1-2.5, while the PDI remained narrow at approximately 2.27. Polymer DF1-1 was synthesized with a medium molecular weight $M_w = 3.7 \times 10^4$ and a narrow PDI of 1.14. Preparation of polymer DF1-25 was not ideal because two components were separated from the polymerization product with different molecular weight 2.2×10^4 and 7.3×10^4 , respectively.

3.2.3. TGA of poly(DMA-co-TFME) polymers

Thermal stability of the polymers was measured by TGA and the curves were showed in Fig. 3. The results indicated that T_d values of the functionalized polymers are $357\text{--}396^\circ\text{C}$ (Table 2). The fastest weight loss of the copolymers began at around 356°C . The copolymers except DF1-2.5 showed a two-step thermal decomposition process. The former weight loss should be due mainly to the decomposition of micromolecules and the later was caused by the degradation of macromolecules with the fracture of chemical bonds. DF1-2.5 showed only one decomposition process due to the extremely small amount of micromolecules residing in the polymer.

T_5 or T_{10} values of the polymers are clearly lower than the corresponding T_d values, causing declining of the TGA curves before the major weight loss of the polymers, which can be attributed to the existence of low-molecular-weight oligomers in the polymers. No distinct regularity was found from T_5 or T_{10} values of the polymers with different TFME content from 3 to 95 wt%. Polymer DF25-1 showed an extremely low T_5 value of 139.5°C , indicating that a slight amount of solvent (e.g. DMF) existed in DF25-1.

Table 2

Molecular weight, PDI, T_d and T_g values of poly(DMA-co-TFME) polymers.

Polymer	TFME wt%	$T_5/^\circ\text{C}$	$T_{10}/^\circ\text{C}$	$T_d/^\circ\text{C}$	$T_g/^\circ\text{C}$	$M_w/\text{g/mol}$	PDI
DF25-1	3.0%	139.5	256.0	395.7	142.4	–	–
DF3-1	20%	256.7	273.7	387.3	98.7	1.25×10^5	7.27
DF1-1	43%	265.9	290.9	393.2	106.5	3.7×10^4	1.14
DF1-2.5	65%	374.7	381.2	384.2	86.6	1.32×10^5	2.27
DF1-10	88%	253.1	285.1	365.9	81.5	–	–
DF1-25	95%	206.3	222.8	357.4	74.2	2.2×10^4 7.3×10^4	1.14 1.12

T_5 : The temperature at which 5% weight loss occurs.

T_{10} : The temperature at which 10% weight loss occurs.

T_d : Thermal-decomposition temperature.

T_g : Glass transition temperature.

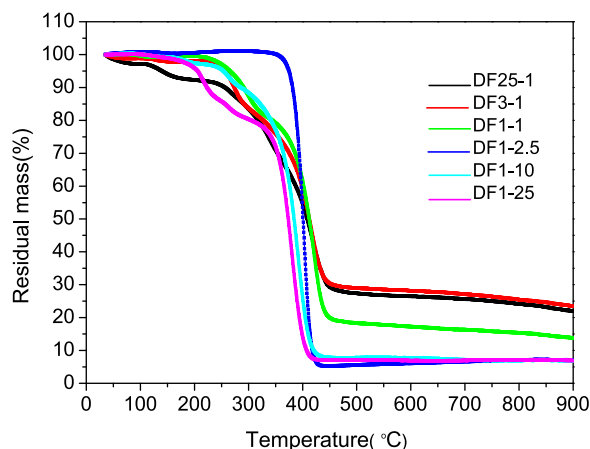


Fig. 3. TGA curves of the functionalized polymers.

It could be observed that T_d values of these polymers decreased with the increase of the content of TFME in the polymers except for DF1-1. On the whole, these polymers showed relatively high T_d values mainly because of their stable C–C skeleton. Introduction of DMA monomer is necessary and it can affect the thermal degradation behavior of the polymers. Dihydroxyphenyl group in DMA monomer imparted high rigidity and thereupon high T_d to the polymers. The high thermal stability of the polymers results from the stable structure of benzene ring and the multiple catecholic groups in the macromolecule chain, which can maintain the binding efficiency by creating multivalent interaction [32]. Thus polymer DF25-1 with the highest content of the rigid structure exhibited the highest T_d value while DF1-25 showed the lowest T_d value. On the other hand, the side chain TFME has little impact on the thermal stability of the polymers though trifluoromethyl group containing C–F bond has a high bond energy [33]. Polymer DF1-1 showed good thermal stability perhaps because the functional groups dihydroxyphenyl and trifluoromethyl groups arranged compactly in the polymerization process.

3.2.4. DSC analysis

DSC measurements were carried out to determine the glass transition properties of the polymers and the curves were showed in Fig. 4. Selection of the operating temperature for DSC measurements on amorphous polymeric materials was determined by T_g values [34]. It is known that in a homogeneous phase system, only one T_g value can be found in the DSC curve [35]. T_g value can be an indicator of the chain mobility or inter-chain interactions [36]. Therefore, the evaluation of T_g will give crucial information regarding the chain interactions.

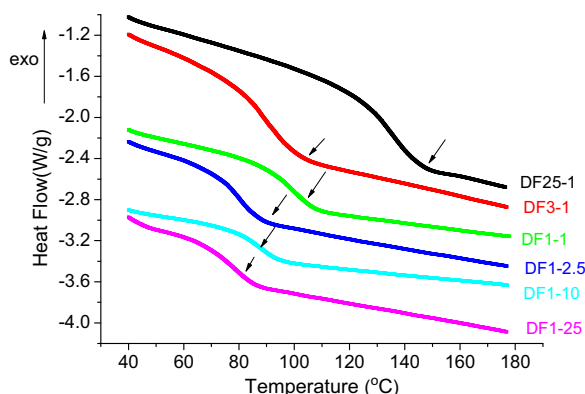


Fig. 4. DSC curves of the polymers.

It could be seen from Fig. 4 that all of the polymers had only one glass transition in the temperature range of 70–150 °C, and no crystallization or melting peak was observed in the high temperature range, reflecting that these polymers were amorphous, which is consistent with the fact that the polymers were obtained by free radical polymerization. The T_g values of the polymers were estimated from their DSC thermograms and were listed in Table 2. It could be found that T_g value decreased from 142.4 °C to 74.2 °C with the increase of TFME content in the polymer from 3 to 95 wt% except DF1-1 (106.5 °C). Generally T_g value of a polymer can be improved by introduction of aromatic rings, polar groups and crosslinking. In the prepared polymers there was strong interaction between the pendant groups due to the intramolecular hydrogen bonding between hydroxyl and carbonyl groups. The size of the pendant group DMA is larger than that of TFME. Therefore there are stronger interactions and larger steric hindrance in polymers with larger DMA content, which conducted to a relatively higher T_g value for polymers with more DMA.

T_g value depends more on the geometry of the constituent groups than on their molecular weights, with the exception of very low molecular weight [37]. Polymer DF1-1 exhibited a higher T_g value though its molecular weight is only 3.7×10^4 g/mol (see Table 2), indicative of the relatively tactic structure and marked steric effect in DF1-1.

3.3. Performance of the polymer films

3.3.1. Adhesion strength and contact angle analysis of the polymer films

Since the solubility of the prepared poly(DMA-co-TFME) copolymers differs greatly in different solvent, they were dissolved in EtOAc or DMF to form solutions of different concentration as showed in Table 3. The solutions were drop-casted onto clean glass, metal or plastic substrates to form transparent films. The ternary polymer DFE could also form transparent solution (50.0 mg DFE in 2 mL of acetone) and was drop-casted onto clean substrates to afford transparent film MDFE.

A protecting or decorating coating must adhere to a substrate for the expected service life. Cross-cut tape test was performed to assess the adhesion of the polymer films to glass, metal or plastic substrates by applying and removing pressure-sensitive tape over cuts made in the films (refer to Test Method: ASTM D 3359-08).

The grid area was inspected for removal of a coating from the substrate or from a previous coating using the illuminated magnifier to rate the adhesion level. Generally coatings of the first three levels (e.g., 0, 1 and 2) are satisfactory for industrial application. As can be seen from Table 4, adhesion strength of the polymer films to metal and plastic substrates was at level 0 or 1. Obviously, the levels (2–4) of coatings adhesion to glass substrate were higher than that to metal and plastic substrates except film MDF1-1, which exhibited the same adhesion strength to glass substrate as that to the other two substrates. Thus the test results indicated that the copolymers were apt to adhere to plastic and metal surfaces. Specially, polymer DF1-1 exhibited potent adhesion strength toward different surfaces. The excellent adhesion strength, especially to metal and plastic substrates of the copolymers could be ascribed to introduction of catechol groups as multivalent hydrogen bonding anchors.

Poly(DMA-co-TFME) films on glass substrate were prepared and their hydrophobicity was studied. Contact angle of deionized water on these films was measured as showed in Fig. 5. Table 4 summarized water contact angle values on surfaces of the synthesized polymer films. The results indicated that the surface contact angles increased gradually with increase of TFME content in the polymer if film MDF1-1 was excluded. The largest contact angle was observed for the film MDF1-25 prepared from polymer

Table 3
Solution of the copolymers for casting films.

Film number	Polymer	Weight/mg	Solvent		Concentration/g/L
			EtOAc/mL	DMF/mL	
MDF25-1	DF25-1	57.3	–	2.0	28.7
MDF3-1	DF3-1	25.0	2.0	–	12.5
MDF1-1	DF1-1	21.2	1.0	–	21.2
MDF1-2.5	DF1-2.5	49.3	2.0	0.5	19.7
MDF1-10	DF1-10	33.1	2.0	–	16.6
MDF1-25	DF1-25	49.3	2.0	–	24.7

Table 4
Water contact angle values and adhesion strength of the polymer films.

Film number	Polymer	TFME wt%	Contact angle H ₂ O film-glass/(°)	Adhesion/level ^a		
				Plastic	Metal	Glass
MDF25-1	DF25-1	3.0%	97.9	0	0	4
MDF3-1	DF3-1	20%	101.5	1	1	3
MDF1-1	DF1-1	43%	111.1	1	1	1
MDF1-2.5	DF1-2.5	65%	105.9	1	1	3
MDF1-10	DF1-10	88%	110.9	1	1	3
MDF1-25	DF1-25	95%	117.4	1	1	2
MDFE	DFE	53%	103.1	1	1	3

0: The edges of the cuts are completely smooth and none of the squares of the lattice is detached.

1: Small flakes of the coating are detached at intersections and less than 5% of the area is affected.

2: Small flakes of the coating are detached along edges and at intersections of cuts. The area affected is 5–15% of the lattice.

3: The coating has flaked along the edges and on parts of the squares. The area affected is 15–35% of the lattice.

4: The coating has flaked along the edges of cuts in large ribbons and whole squares have detached. The area affected is 35–65% of the lattice.

5: Flaking and detachment are worse than level 4.

^a Adhesion strength of the polymer films is divided to six levels (0–5) as follows:

DF1-25 with the highest fluorine content, indicating its strongest hydrophobicity. The least hydrophobic polymer in the series was DF25-1 with a contact angle of 98° and others showed intermediate values, wherein DF1-25 and DF1-10 showed the similar and good water resistance. The contact angle experiments demonstrated that the introduced trifluoromethyl groups aggregated on the outer surface in the polymers and the hydrophobicity of the copolymers could be fine tuned by altering the ratio of reactive monomers.

3.3.2. SEM images of the polymer films

SEM images of cross-section and skin layer of the prepared films with different TFME contents were measured and showed in Fig. 5. Surfaces of most films such as MDF25-1, MDF1-2.5 and MDF3-1 appeared relatively flat. Film MDF1-10 exhibited more porosity and larger macrovoids compared to other polymer films. Increase of TFME content in the polymer enhances the

hydrophobicity of the polymeric casted films. Meanwhile the interaction between polymer and solvent molecules may decrease as a result of obstruction derived from the rigid structure of benzene ring and trifluoromethyl groups. The polymer chains diffuse between solvent layers, which weakens the interaction between the polymer chains and solvent and results in increasing of solvent out flow [38,39]. Therefore the porosity of film MDF1-10 is larger and its skin layer thickness is smaller than that of other poly(DMA-co-TFME) films, respectively.

Polymer DF1-2.5 with 65 wt% of TFME content formed films with thicker and denser skin layer and denser structure (less porosity) as showed in Fig. 6(a). Different feed molar ratios between two different monomers for preparation of copolymers may affect the mechanism of film formation in the solvent and, consequently, the smoothness of film surfaces as well as the compactness of the films [38]. When EG9ME (the third monomer) was introduced into the

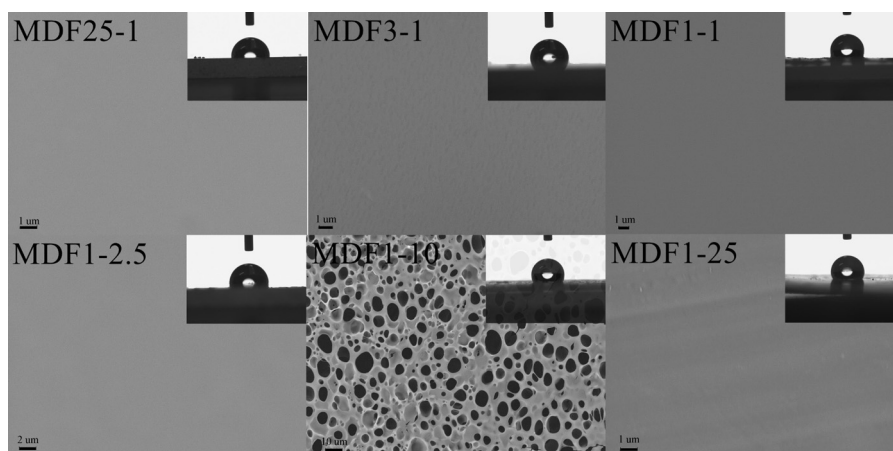


Fig. 5. SEM images of the upper surfaces of poly(DMA-co-TFME) films and the insets are the images of their contact angles on the surfaces of the copolymer.

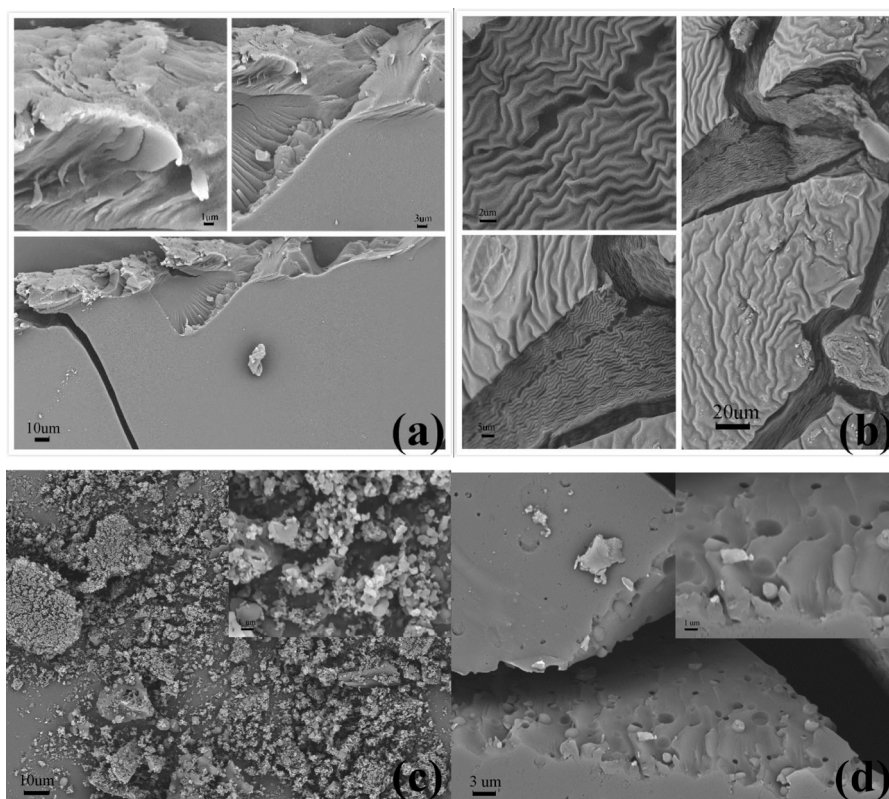


Fig. 6. SEM images of (a) membrane of polymer DF1-2.5 and its cross-sectional view; (b) the cross-sectional structure of the prepared terpolymer film; (c) copolymer DF1-1 (the inset is the magnified image at 10,000 magnification); and (d) surface and cross-section of membrane MDF1-1 (the inset is the magnified image at 10,000 magnification).

polymer chain a terpolymer was obtained and SEM images of the cross-sectional structure of this terpolymer film were showed in Fig. 6(b). The fracture surface of the film is plicated and its cross-section presents inner wrinkled shape. This can be attributed to addition of PEG with soft characteristics and decrease in the hardness of the terpolymer, which has excellent film-forming property and is viscous and easy to process. SEM images of polymer DF1-1 and its film MDF1-1 are showed in Fig. 6(c) and (d), respectively. It can be found that amorphous particles emerge in the polymer and the diameter of the small particles is about 100 nm. There are some pores of varied size diffusing on the film MDF1-1 surface and its internal part. The results suggest that the prepared copolymer films have potential to be developed to biomimetic anti-biofouling coatings.

3.3.3. Test on resistance to chemicals of the polymer films

The formed polymer films were immersed in distilled water for 7 days till their weights were constant so as to calculate their water absorption and thereby to evaluate their hydrophilicity. Test

results on water absorption and resistance of polymer films to various chemicals were listed in Table 5. The measured film MDF1-25 was easiest to absorb water while film MDF3-1 hardly absorbed moisture, showing good hydrophobicity. Slime, algae and bryozoa are commonly found in low-salinity waters and these species are sources of pollutions from ships. Therefore, it is necessary to assess the resistance to salt water of antifouling coatings used on marine ships. After being immersed into salt water for one month, poly(DMA-co-TFME) films displayed no change and their surfaces still closely adhered to the substrate. All of the polymer films did not fall off from the substrate after immersion in 20% sodium hydroxide solution for seven days and the surface materials did not dissolve, indicative of good alkali resistant property of the synthesized copolymers. However, it was found that a very small amount of the film materials dissolved away after the polymer films were immersed in 20% NaOH-H₂O/EtOH (10% volume of EtOH) solution for seven days, revealing relatively a little weak solvent resistance of the polymers. Most of the polymer films also exhibited good resistance to 20% dilute sulfuric acid. In conclusion, the

Table 5
Resistance of polymer films to various chemicals^a.

Membrane number	5 wt% saline solutions (1 month)	20% dilute H ₂ SO ₄ (7 days)	20% NaOH solution (7 days)	20% NaOH solution in H ₂ O/EtOH (7 days)	water absorption (%)(7days)
MDF25-1	good	general	good	good, slight change	0.33%
MDF3-1	good	good	good	good, slight change	0.05%
MDF1-1	good	good	good	good, slight change	2.20%
MDF1-2.5	good	good	good	good, slight change	0.89%
MDF1-10	good	good	good	good, slight change	0.13%
MDF1-25	good	Not good	good	good, slight change	2.31%

^a "Good" means that the film did not fall off from the substrate.

Slight change: a very small amount of the film materials was dissolved.

^a Water absorption was calculated based on the equation: water. Absorption = $(M_1 - M_2)/M_2 \times 100\%$, wherein M_1 is weight of the wet film after 7 days and M_2 represents weight of the film before being immersed in water.

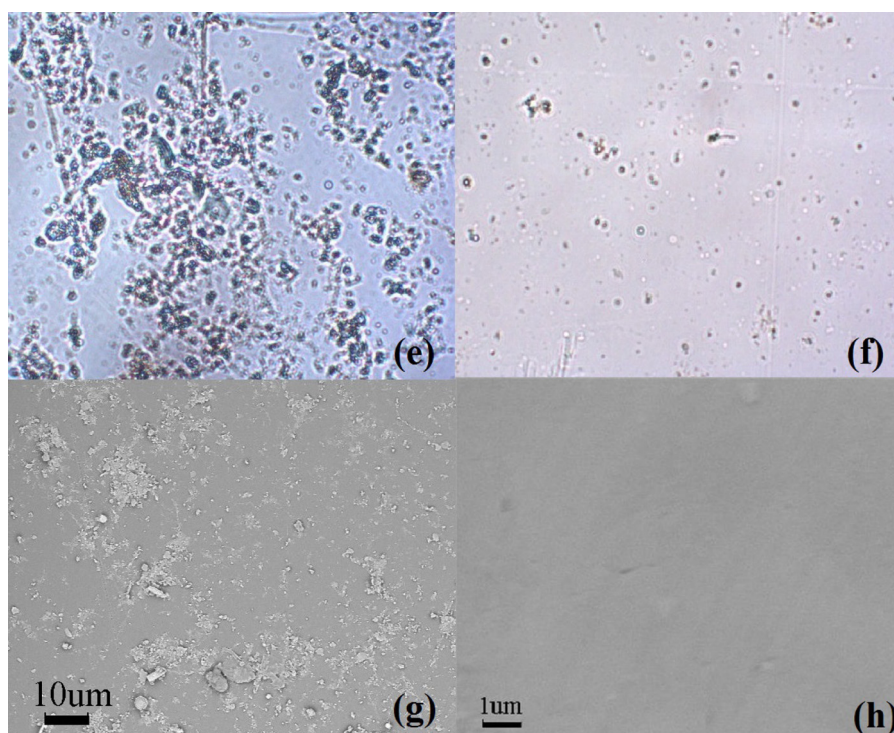


Fig. 7. Micrograph image of the surface of the blank glass slide (e) and the coatings of polymer DF1-1 (f) that were immersed in seawater for seven days and SEM image of the blank glass slide (g) and the coatings of polymer DF1-1 (h) after immersion in seawater for seven days.

experimental results on resistance to chemicals corrosion demonstrated that the polymer films showed good resistance to water, salt, acid and alkali.

3.3.4. Antifouling performance test

Generally, the biofouling started from the adhesion of bacteria or algae on the surfaces immersed in natural water [40]. Therefore blank glass slides and the obtained copolymer coatings on glass were used to examine anti-biofouling property of the polymers. The blank slides and the coatings of polymer DF1-1 on glass were immersed in seawater containing various microbes for seven days. The microbial adhesion on the slides and the coatings was showed in Fig. 7. The hydrophilic surface of the blank glass slide was seriously biofouling after one week as could be observed from Fig. 7(e) and (g). For the panels coated with polymer DF1-1 less fouling on the surface was observed as expected (Fig. 7(f) and (h)). It was proved that the copolymer coatings could inhibit biofouling to some extent. Potent antimicrobial adhesion could be achieved by introduction of TFME to the block copolymer because trifluoromethyl groups endowed both antifouling property and low surface energy. The fluorine-containing polymer coatings are free from heavy metals and consequently will not give rise to electrochemical corrosion and heavy metal pollution. The green polymer coatings containing DMA and TFME or EG9ME segments can degrade into oligomers by cleavage of the ester bond and finally decompose into carbon dioxide and water in the environment. Reasonably, permanent antifouling performances might be expected for the polymer films on account of their strong adhesion to substrates, which will be evaluated in depth in the near future.

4. Conclusions

In the current work, new functionalized copolymers containing catechol and trifluoromethyl groups were prepared by radical polymerization. The polymers displayed excellent thermostability.

Catechol groups can anchor the copolymers onto surfaces of a wide variety of materials and trifluoromethyl groups impart the film surface free energy on flat surfaces and excellent hydrophobicity on a slice of rough surfaces. The strong adhesive force of the catechol-containing polymers is plausibly due to the multivalent chelating interaction (in the case of metals) and/or hydrogen bonding (in the case of nonmetals such as plastic and glass). Results of the tests including FT-IR spectra, GPC, TGA, DSC and SEM of the polymers and the polymer films led us to the conclusion that copolymer DF1-1 (with a TFME content of 43%) would be a preferential selection on account of both internal structure and superior performance of its films, including excellent resistance to water, salt, acid and alkali and fair antifouling property. Moreover, EG9ME, with good antibacterial property, can be used as the third monomer and incorporated into the copolymers so as to improve the performance of the polymer antifouling coatings. The starting materials used for synthesis of the polymers are inexpensive and easily available. The copolymers are readily to degrade and environmentally friendly as a result of short trifluoromethyl pendant side groups. Thus the polymer films reveal tremendous potential as promising marine antifouling coatings. Nevertheless, powerful fouling resistance of delicate surface structures necessary to maintain super amphiphobicity continues to be a challenge. Current and ongoing efforts in our laboratory continue to address the challenge by using simple and scalable fabrication approaches.

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